

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.70%	Share of actual attacks successfully flagged
F1-Score	98.84%	Balanced measure combining Precision and Recall
Accuracy	99.68%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.30%	Actual attacks that went undetected
True Positive Rate (TPR)	97.70%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 11089	FN 261
Actual: Normal	FP 0	TN 69752

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	11089	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Response Latency

Metric	Value	Description
Detection Time	498.4 ms	Average time taken to detect an attack
Mitigation Response Time	4.9 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.46%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	9.45%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	10.26%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	8	3	2026-09-11 10:46:51	2026-09-11 11:00:50
10.0.0.17	7	3	2026-09-11 10:46:52	2026-09-11 10:59:31
10.0.0.18	7	3	2026-09-11 10:46:59	2026-09-11 10:58:38
10.0.0.19	6	2	2026-09-11 10:46:54	2026-09-11 10:59:35

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-11 10:46:51	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:46:52	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:46:54	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:46:59	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:47:02	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (1m)
2026-09-11 10:47:03	10.0.0.17	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-11 10:47:03	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-11 10:47:04	10.0.0.17	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-11 10:47:05	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-11 10:47:05	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-11 10:47:10	10.0.0.18	DDoS	SYN Flood	91.4%	Critical	Time Ban (1m)
2026-09-11 10:47:10	10.0.0.18	DDoS	SYN Flood	91.4%	Critical	Time Ban
2026-09-11 10:48:02	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (2m)
2026-09-11 10:48:03	10.0.0.17	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)
2026-09-11 10:48:05	10.0.0.19	DDoS	SYN Flood	93.9%	Critical	Time Ban (2m)
2026-09-11 10:48:10	10.0.0.18	DDoS	SYN Flood	91.4%	High	Released
2026-09-11 10:48:20	10.0.0.18	DDoS	SYN Flood	93.8%	Critical	Time Ban (2m)
2026-09-11 10:58:38	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Blackhole
2026-09-11 10:59:30	10.0.0.16	DDoS	SYN Flood	93.2%	Critical	Blackhole
2026-09-11 10:59:31	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Blackhole
2026-09-11 10:59:35	10.0.0.19	DDoS	SYN Flood	94.3%	Critical	Released
2026-09-11 10:59:35	10.0.0.19	DDoS	SYN Flood	94.3%	Critical	Blackhole

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.16	SYN Flood	0.6341	92.5%	Critical	Phase 3	1	1s	2026-09-11 11:00:50	Escalated to Blackhole
10.0.0.19	SYN Flood	0.6402	94.3%	Critical	Phase 2	1	1m 11s	2026-09-11 10:59:35	Ban Expired
10.0.0.17	SYN Flood	0.6271	92.7%	Critical	Phase 2	2	1m 11s	2026-09-11 10:59:31	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6176	92.7%	Critical	Phase 3	3	1m 11s	2026-09-11 10:59:31	Escalated to Blackhole
10.0.0.16	SYN Flood	0.6275	93.2%	Critical	Phase 2	2	1m 10s	2026-09-11 10:59:30	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6275	93.2%	Critical	Phase 3	3	1m 10s	2026-09-11 10:59:30	Escalated to Blackhole

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.18	SYN Flood	0.6427	90.9%	Critical	Phase 2	1	11s	2026-09-11 10:58:38	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6392	90.9%	Critical	Phase 3	2	11s	2026-09-11 10:58:38	Escalated to Blackhole
10.0.0.19	SYN Flood	0.6516	91.9%	Critical	Phase 2	1	11s	2026-09-11 10:58:35	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6260	92.7%	Critical	Phase 2	1	11s	2026-09-11 10:58:31	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6380	93.2%	Critical	Phase 2	1	10s	2026-09-11 10:58:30	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6067	92.7%	Critical	Phase 2	2	1m 11s	2026-09-11 10:54:18	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6466	92.7%	Critical	Phase 2	2	1m 10s	2026-09-11 10:54:12	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6281	92.9%	Critical	Phase 2	2	1m 11s	2026-09-11 10:54:10	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6327	91.0%	Critical	Phase 2	2	1m 10s	2026-09-11 10:54:10	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6473	92.7%	Critical	Phase 2	1	11s	2026-09-11 10:53:18	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6474	92.7%	Critical	Phase 2	1	10s	2026-09-11 10:53:12	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6371	92.9%	Critical	Phase 2	1	11s	2026-09-11 10:53:10	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6327	91.0%	Critical	Phase 2	1	10s	2026-09-11 10:53:10	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6390	93.8%	Critical	Phase 2	3	10s	2026-09-11 10:48:20	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.5983	91.4%	High	Phase 2	1	1m 11s	2026-09-11 10:48:10	Ban Expired
10.0.0.19	SYN Flood	0.6413	93.9%	Critical	Phase 2	2	1m 11s	2026-09-11 10:48:05	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6191	93.1%	Critical	Phase 2	2	1m 11s	2026-09-11 10:48:03	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6277	92.9%	Critical	Phase 2	2	1m 11s	2026-09-11 10:48:02	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.18	SYN Flood	0.6422	91.4%	Critical	Phase 2	1	11s	2026-09-11 10:47:10	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6473	90.9%	Critical	Phase 2	1	11s	2026-09-11 10:47:05	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6259	93.1%	Critical	Phase 2	1	11s	2026-09-11 10:47:03	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6383	92.9%	Critical	Phase 2	1	11s	2026-09-11 10:47:02	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE